Deliverable 3 – Dataset Understanding & Data Model Report

Healthcare Patient Experience Analytics using Power BI

Introduction

This report provides an overview of the datasets used in the **Healthcare Patient Experience Analytics** project. Understanding the structure and relationships between the datasets is essential for building an efficient Power BI data model and producing reliable business insights.

The project uses multiple datasets containing healthcare survey information collected through the **Hospital Consumer Assessment of Healthcare Providers and Systems (HCAHPS)** survey. These datasets include information about healthcare quality measures, survey questions, reporting periods, state information, national results, state results, and survey response categories. The datasets are organized using a **Star Schema** to optimize reporting performance and simplify analytical queries.

Dataset Overview

The project consists of the following datasets:

Table Name	Table Type	Description
Measures	Dimension	Stores healthcare quality measures evaluated through the HCAHPS survey.
Questions	Dimension	Contains patient survey questions related to each quality measure.
Reports	Dimension	Stores reporting period information used for trend analysis.
States	Dimension	Contains state names, state codes, and regional classifications.
National Results	Fact	Stores national-level patient

Table Name	Table Type	Description
		satisfaction results.
State Results	Fact	Stores patient satisfaction results for individual states.
Responses	Fact	Stores survey response categories and percentage values used for KPI calculations.

Table Details

1. Measures Table

Purpose

Stores the healthcare quality measures evaluated in the HCAHPS survey.

Business Description

The table defines different aspects of patient experience such as communication with doctors, communication with nurses, hospital cleanliness, pain management, and discharge information. These measures allow healthcare organizations to evaluate the quality of healthcare services.

Primary Key

- Measure ID

Important Columns

Column	Description
Measure ID	Unique identifier for each healthcare measure
Measure Name	Name of the healthcare quality measure
Measure Description	Detailed description of the measure

2. Questions Table

Purpose

Stores the survey questions answered by patients.

Business Description

Each healthcare quality measure contains one or more survey questions. This table allows detailed analysis of patient responses for every healthcare quality dimension.

Primary Key

- Question ID

Important Columns

Column	Description
Question ID	Unique identifier for each survey question
Question Text	Survey question presented to patients
Measure ID	Links each question to its healthcare quality measure

3. Reports Table

Purpose

Stores reporting period information.

Business Description

This table enables comparison of healthcare performance across different reporting periods and supports trend analysis.

Primary Key

- Report ID

Important Columns

Column	Description
Report ID	Unique reporting period identifier
Reporting Period	Survey reporting period
Start Date	Beginning of reporting period
End Date	End of reporting period

4. States Table

Purpose

Stores geographical information for participating states.

Business Description

The table enables state-level and regional healthcare performance analysis by linking survey results with state and regional information.

Primary Key

- State Code

Important Columns

Column	Description
State Code	Unique state identifier
State Name	Name of the state
Region	Region to which the state belongs

5. National Results Table

Purpose

Stores national patient satisfaction results.

Business Description

This table contains aggregated healthcare survey results at the national level and serves as the benchmark for comparing state performance.

Primary Key

- National Result ID *(or composite business key, depending on the dataset structure)*

Important Columns

Column	Description
Report ID	Reporting period
Measure ID	Healthcare quality measure
Question ID	Survey question
Response ID	Survey response category
Percentage	National satisfaction percentage

6. State Results Table

Purpose

Stores patient satisfaction results for every participating state.

Business Description

This is the primary analytical dataset used for state performance analysis, regional comparison, rankings, and trend analysis.

Primary Key

- State Result ID *(or composite business key, depending on the dataset structure)*

Important Columns

Column	Description
State Code	State identifier
Report ID	Reporting period
Measure ID	Healthcare quality measure
Question ID	Survey question
Response ID	Survey response category
Percentage	State satisfaction percentage

7. Responses Table

Purpose

Stores survey response categories.

Business Description

The table categorizes survey responses into **Top-box**, **Middle-box**, and **Bottom-box** percentages, which are used to calculate patient satisfaction KPIs.

Primary Key

- Response ID

Important Columns

Column	Description
Response ID	Unique response category identifier
Response Category	Top-box, Middle-box, or Bottom-box
Response Description	Description of the response category

Fact Tables

The following tables are classified as **Fact Tables** because they store measurable healthcare survey data.

Fact Table	Business Purpose
National Results	Stores national patient satisfaction metrics used for benchmarking and KPI calculations.
State Results	Stores state-level patient satisfaction metrics used for regional and state performance analysis.
Responses	Stores response category values used in percentage calculations and KPI reporting.

These tables contain quantitative values that are analyzed across multiple business dimensions.

Dimension Tables

The following tables are classified as **Dimension Tables** because they provide descriptive information for filtering and grouping data.

Dimension Table	Business Purpose
Measures	Describes healthcare quality measures.
Questions	Describes patient survey questions.
Reports	Describes reporting periods.
States	Describes state and regional information.

Dimension tables provide the business context required for meaningful analysis and interactive dashboard filtering.

Star Schema Design

The project follows a **Star Schema** architecture to improve reporting performance and simplify analytical queries.

Benefits of the Star Schema

- Reduces data redundancy.
- Improves Power BI query performance.
- Simplifies DAX calculations.
- Supports interactive filtering.
- Provides a scalable and maintainable data model.

In this design, the **National Results**, **State Results**, and **Responses** tables act as the central fact tables, while **Measures**, **Questions**, **Reports**, and **States** function as dimension tables connected through one-to-many relationships.

Relationship Diagram

The following relationships are implemented within the Power BI data model:

From Table	To Table	Relationship
Measures	National Results	One-to-Many
Measures	State Results	One-to-Many
Questions	National Results	One-to-Many
Questions	State Results	One-to-Many
Reports	National Results	One-to-Many
Reports	State Results	One-to-Many
States	State Results	One-to-Many
Responses	National Results	One-to-Many
Responses	State Results	One-to-Many

These relationships ensure accurate filtering, aggregation, and drill-down across all dashboard visualizations.

Data Model Summary

The implemented data model supports:

- National healthcare performance analysis
- State-wise patient satisfaction analysis
- Regional performance comparison
- Patient experience measure analysis
- Reporting period trend analysis
- National benchmarking
- Interactive Power BI dashboards
- KPI calculations using DAX
- Efficient analytical reporting through a Star Schema

Conclusion

The Healthcare Patient Experience Analytics project uses a well-structured Star Schema consisting of fact and dimension tables. This data model enables efficient dashboard development, improves report performance, and provides a reliable foundation for KPI calculations, benchmarking, trend analysis, and executive decision-making. The structured relationships between datasets ensure accurate analysis of patient satisfaction across different healthcare dimensions and geographical regions.
